

Notes:

H gate works on **one** qubit to create mystery and probability (superposition).

CX gate works on **two** qubits to create a link or a rule between them (entanglement).

AI Chats: Claude, ChatGPT

-google colab

NISQ: Noisy Intermediate-Scale Quantum. It is the name for the current generation of quantum computers because they are noisy, today's quantum computers can only run **short**, quick calculations before the errors take over and ruin the results.

-five types of quantum noise models

Depolarizing Error: Random errors (X(bit flip), Y(bit+phase flip), or Z(phase flip))

- Depolarizing noise caused the Bell state to become less accurate by introducing random errors. While most measurements remained the expected **00** and **11**, a small number of **01** and **10** outcomes appeared. These incorrect outcomes show that the entangled state was partially disrupted by the depolarizing noise.--> placed after the CX gate

error = depolarizing_error(**1.0**, 1)

- Output: {'11': 50100, '00': 49900}
- Even with the maximum amount of noise injected, there is still no noticeable impact.
- At $\lambda=1$, Qiskit's depolarizing formula reduces to $E(\rho) = I/2$ — full noise on H doesn't scramble the qubit, it just replaces it with a plain random coin flip. CX then copies that random result perfectly, so you still get 50% 00 / 50% 11 — same counts as the ideal Bell state. Your measurement only checks "do the two qubits agree," and both the real entangled state and this noisy fake version agree 100% of the time. So counts can't distinguish them, no matter how much noise you add to H.
- CX noise is different: it corrupts the copying step itself, so agreement breaks (you get 01/10), which counts *can* detect — even at small p.

- The code runs the same Bell circuit twice, once measuring normally (Z basis) and once with an extra H added to both qubits right before measurement (X basis). In the Z-basis run, ideal and noisy circuits look identical — both give 50/50 on 00/11 — because that measurement only checks whether the two qubits agree, and full H noise doesn't break agreement, it just removes the "quantum-ness" behind it. The X-basis run is the fix: it checks something noise actually damages — coherence — instead of just agreement. There, the ideal circuit stays perfectly correlated, but the noisy one collapses into pure randomness (25% each outcome), proving the noise was real all along.

NEW: Depolarizing noise for the CX gate

`cx_error = depolarizing_error(0.10, 2)`

`noise_model.add_all_qubit_quantum_error(cx_error, ['cx'])`

- Adding noise to the CX gate of 0.10 → to get output.
- {'01': 2474, '11': 47282, '10': 2572, '00': 47672}
- Even adding noise of 0.01 detects error

Try

- Bit-flip noise
- Phase-flip noise
- Amplitude damping
- Phase damping
- Thermal relaxation

Bit Flip Noise: when the value of a digital bit changes from 0 to 1, or from 1 to 0: Accidental X gate

- Added a 0.05 error to one H gate: result showed no error

- Added a 0.05 error to one CX gate: result, The bit-flip noise introduced errors into the entangled state.
 - ❖ The "bit-flip error" is like someone reaching in and swapping the labels "heads" and "tails" on that spinning coin. But if the coin is already perfectly balanced between both, swapping the labels doesn't actually change anything — it's still the exact same spinning coin, indistinguishable before and after. That's why even a guaranteed (100%) bit-flip on H does nothing: the state H creates is already symmetric under that exact flip, so the "error" has nothing to disturb.

Contrast that with CX noise: by that point the qubits aren't a balanced coin anymore — CX has already linked them into a specific outcome (like "if this one's heads, that one's tails"). A bit-flip there is swapping a value that's already been pinned down, so it genuinely changes the result you'll see. That's why it shows up immediately, even at just 5%.

Phase Flip: a 180-degree rotation of a quantum bit's state: Accidental Z Gate

- Added a phase flip error on H gate → results showed no error.
- Added a phase flip error on CX gate → results showed no error.
 - ❖ Phase flip inverts the sign of the base values, not the qubits logical values like bit flip. (does not hold the power to be able to change anything)
- If we wanted error to show, we would have to change how we measure and not the noise-->

The code runs the circuit twice, once in z basis and once in the x basis. the noise model, the h gate, the cx gate, and the error itself doesn't change. but in the x basis pass: the qubits final state gets asked are you 0 or 1. In the x basis pass you rotate the qubits and then ask the 0 or 1. because of the rotation that revealed its damage. → result shows error.

Amplitude Damping: the process where a tiny quantum bit (qubit) loses its energy to the outside world and drops from an active high state down to a resting low state. [1]
Think of a qubit like a light switch or a tiny spinning top.

When it has energy, it is in an "excited" state (like state 1).

When it loses that energy, it falls into a "ground" or resting state (like state 0).

This drop is a one-way street. The qubit loses energy to its environment, but it does not get that energy back on its own. [1, 2]

- Added an amplitude damping on H gate → results showed error with a skew of values vs 5000-5000
- Added an amplitude damping on CX gate → results showed error.

- ❖ Random skew flips sides across runs → that's just shot noise. A real error skews the same direction every time. Phase-flip's wobble was noise; amplitude damping's skew was real.
- ❖ On H, the qubit comes out as a clean 50/50 blend of 0 and 1, plenty of "1" population for damping to eat into, so the balance visibly shifts. On CX, the qubits are already correlated pairs (one of them is always 1 in your Ψ^+ state), damping can knock either of those 1s down to 0, producing an outcome (00) that shouldn't exist at all.
- ❖ Contrast that with bit-flip and phase-flip: both are symmetric operations that can't tip the 0/1 balance, so they had nothing to exploit on H specifically (H's output happens to be exactly the symmetric point they can't disturb). Amplitude damping has no such blind spot, any qubit with 1-population is fair game, on any gate, which is why it shows up on both.

Phase Damping: qubit gradually losing its phase coherence because it interacts with its environment. Initially the $|0\rangle + |1\rangle$ have a defined relationship, but as time passes the environmental interactions make the relationship less precise. Eventually the superposition is lost, even though the probabilities of measuring 0 or 1 haven't changed.

- Applied Phase Damping Error to Hadamard Gate → Z-basis measurement isn't very sensitive to the coherence loss caused by phase damping.
 - ❖ Phase damping on the H gate did not noticeably change the measurement results because I measured the circuit in the Z basis. Phase damping does not change the measured bits (01 and 10); instead, it weakens the quantum coherence (the phase relationship) between them. Since the Z-basis measurement cannot directly detect this loss of coherence, the output looked almost the same as the ideal case.
- Applied phase damping error to CX gate → no error was detected.
 - ❖ Phase damping on the CX gate also produced almost the same measurement counts as the ideal circuit. Although the noise reduced the quantum coherence of the Bell state, it did not change the probabilities of measuring 01 and 10 in the Z basis. As a result, no obvious errors appeared in the measurement counts, even though the quantum state was affected internally.

